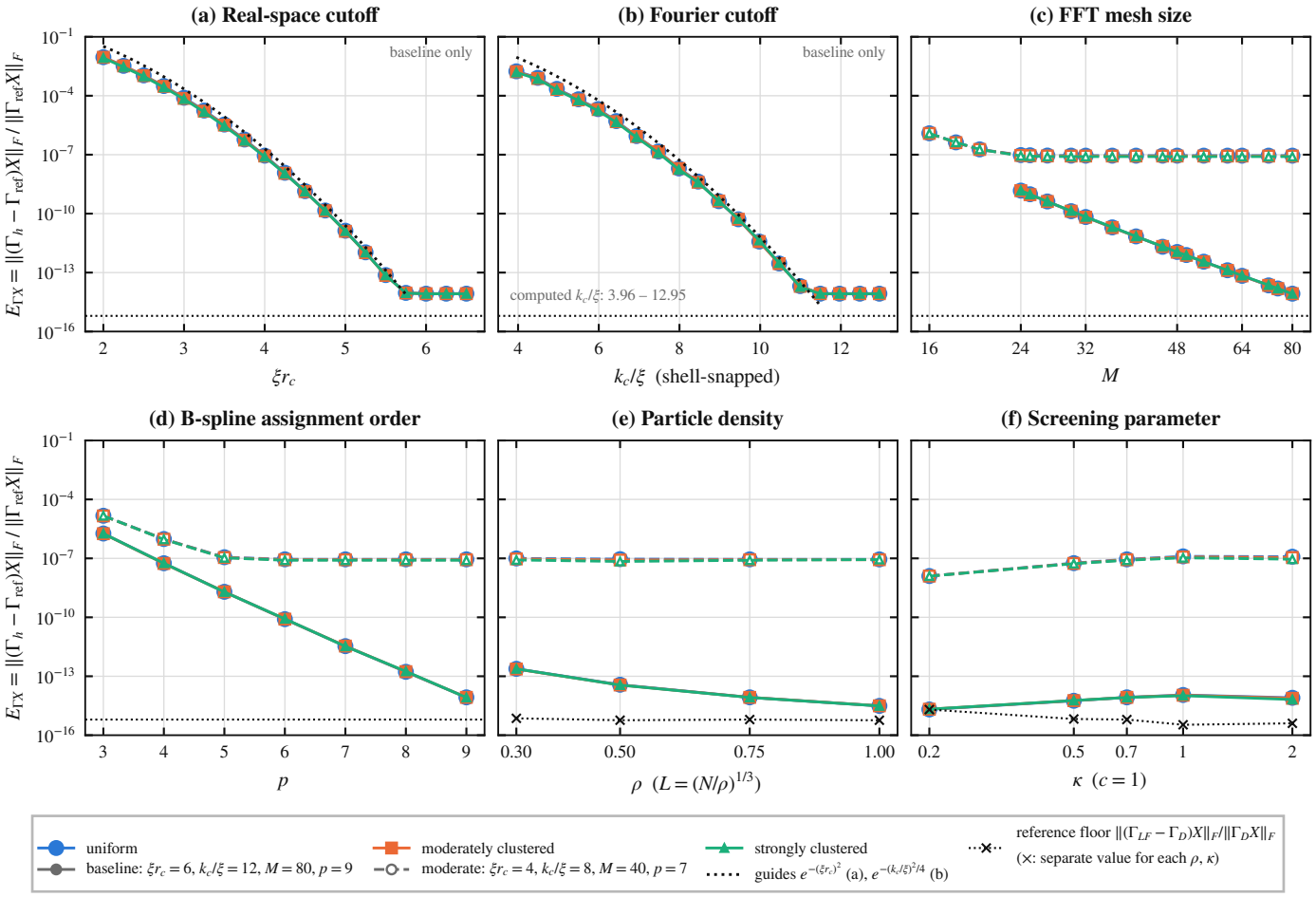
(a) Real-space cutoff
(b) Fourier cutoff
(c) FFT mesh size
(d) B-spline assignment order
(e) Particle density
(f) Screening parameter
$E_{TX} = \|(\Gamma_h - \Gamma_{\mathrm{ref}})X\|_F / \|\Gamma_{\mathrm{ref}}X\|_F$
baseline only
baseline only
computed k_c/ξ: 3.96 – 12.95
ξr_c
k_c/ξ (shell-snapped)
M
p
ρ $(L = (N/\rho)^{1/3})$
κ $(c = 1)$
uniform
baseline: $\xi r_c = 6$, $k_c/\xi = 12$, $M = 80$, $p = 9$
moderately clustered
moderate: $\xi r_c = 4$, $k_c/\xi = 8$, $M = 40$, $p = 7$
strongly clustered
guides $e^{-(\xi r_c)^2}$ (a), $e^{-(k_c/\xi)^2/4}$ (b)
reference floor $\|(\Gamma_{LF} - \Gamma_D)X\|_F / \|\Gamma_D X\|_F$
(×: separate value for each ρ, κ)